TUTOR GUIDE • LEVEL 1 • TERM 2 • LESSON 10 OF 10

# Find One First




**Big idea:** Find one, then find many. When we know what a group costs, we can find what one costs by sharing equally. Then we can find what any number of them cost.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

## 1. At a glance

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| **Strand** | Multiplication and Division; Money (Multiplication and division word problems: divide within 20, multiply within 40 (Primary 1, Multiplication and Division, p. 31); Count money in dollars (Primary 1, Money, p. 31): whole dollars only, no notes, coins or cents until Term 3)<br>Strand label for the tutor only. This is an added lesson (gap from the word-problem books analysis: the price or size of one from a group). Start here is Primary 1 content: share a total equally to find one (within 20), then multiply to find many (within 40), in whole dollars. BEYOND PRIMARY 1 ON PURPOSE: Think harder and Go further chain the two steps with different numbers (the price of 7 from the price of 3), use the words a dozen and such, ask how many can be bought, and compare the price of one in two shops. Owner decision, 22 Oct 2026: strong content with scaffolding, said plainly to families. Say this on the parent page and in the opening of the guide; do not describe the lesson as MOE-aligned or syllabus-aligned. Money is only whole dollars and $1 tokens (counters): no notes, coins or cents until Term 3. Vocabulary taught here: each, a dozen, such. This lesson also holds Mi's Challenge Quest (20 minutes, minutes 35 to 55; a game with Mi, not a test); the tutor notes with the private notes sheet and the grown-up note are in the Quest files built by assets/quest_build.py. Contest tag (tutor only): price of one first, then many; how many can be bought; compare the price of one. |
| **Objectives** | • Find the cost of one by sharing a total equally ("4 pens cost $20, so 1 pen costs $5") and the cost of many from the cost of one ("1 pen costs $3, so 4 pens cost $12").<br>• Draw a strip of equal parts for the story and write both steps: find one first, then find many.<br>• Use the words each, a dozen and such.<br>• Think harder: find the price of 7 from the price of 3, use a dozen and such, and say how many can be bought.<br>• Go further: one price from each of two groups, money left over, and which shop sells one for less.<br>• Play Mi's Challenge Quest, a game with Mi on private paper: 8 short gates that visit the lessons of the term. |
| **Joining cold** | **Remember warm-up covers:** 12 counters shared equally onto 3 plates (Term 2 Lesson 7, taught again with plates); Each and in all: 3 bags with 4 counters in each bag (Term 2 Lessons 5 and 8, taught again with a picture); A strip of 4 equal parts with the whole strip 20 (Term 2 Lesson 8, taught again with a picture).<br>Only counting to 20 and adding small numbers. Nothing from an earlier lesson is needed: the warm-up deals counters onto plates, shows bags with the same number in each bag and shows a strip cut into equal parts. The words each, a dozen and such, and the idea of a price, are taught in Explore and the Mini-lesson.<br>Try not to miss this lesson: it holds Mi's Challenge Quest. A child who misses it plays at the make-up slot (small group, a parent beside the child on camera) BEFORE watching the recording: the quest is not in the recording. The lesson itself builds on Term 2 Lessons 5 to 8 (groups, tables, sharing, stories), and a child who joins here without them can follow the warm-up and the first workbook page; the line for the grown-up covers only what the child met, and the family may leave out gates for lessons the child did not attend. |

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| **Child materials** | • The printed workbook with the $1 tokens kit page (page 6): a grown-up cuts out the 30 tokens beforehand (or use about 30 counters, beans or buttons as $1 tokens; the lesson needs at most 28)<br>• 3 to 6 small things to be the shop items (pencils, rubbers, cups or toys)<br>• The Quest Log, printed at 100% and kept folded or closed in an envelope (sent a week before, with the grown-up note), a quest partner (a grown-up or a brother or sister), and a pencil and rubber<br>• A pencil, a rubber and spare paper |
| **Tutor materials** | • The slide deck and the quest files (tutor notes with the private notes sheet, grown-up note)<br>• A few shop items and some tokens to show on camera if you wish<br>• Your own watch or clock, out of sight of the camera. No timer or clock is shown during the quest<br>No ruler is needed. Nothing needs to be bought: everything can be found at home, and the tokens can be counters, beans or buttons. No matches, toothpicks or anything sharp. Small things never go near the mouth or ears; keep them away from younger brothers and sisters. No real money is needed and no notes or coins are shown. |
| **Tiers** | Start here (6 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on. |
| **Contest-style move** | Used at Think harder only: find the price of one first, then the price of many; how many can be bought with a given amount; compare the price of one in two shops. Never advertised to children or parents. |

**Before the lesson**

- Solve every workbook question and every quest gate (and the secret cave item) yourself first, with the quest tutor notes (L20_Quest_Tutor_Notes.pdf): the answers, the private notes sheet for each child and the group advice sheet. Money is whole dollars only; do not bring notes or coins into the lesson.
- A week before, send the families the grown-up note (L20_Quest_Parent_Note.pdf), the Quest Log (L20_Quest_Log.pdf, 3 pages, print at 100%) and the lesson materials: the workbook with the tokens kit page (a grown-up cuts out the 30 tokens), a few small shop items, and a closed envelope for the Quest Log. A family with no printer tells you: arrange the make-up slot. Build the quest files with: bash term/l1/build.sh 20 then python3 term/l1/assets/quest_build.py 20.
- Have the slides open. The quest is straight after the brain break, minutes 35 to 55, then the workbook. Have the quest tutor notes ready and your own watch where only you can see it. Never show a clock, countdown or timer on the shared screen during the quest. Delete each child's notes sheet when the line is sent.
- Check the recording captures the tutor and the shared screen. Children video is excluded; any unmuted voice is recorded. Greet each child by first name as they arrive, then start the recording, then say the notice; avoid first names while it runs. PAUSE the recording for the whole quest (minutes 35 to 55) and restart it for the workbook. The business WhatsApp number is never said or shown on a slide; it is in the grown-up note.
- Be plain with families: sharing within 20 and multiplying within 40 in whole dollars is Primary 1 work, but chaining the steps with different numbers, a dozen, such and how many can be bought go beyond the school Primary 1 syllabus on purpose. Do not call the lesson MOE-aligned. Use the words shared equally, not a division sign. The tokens and the strip are the method, not a rule to recite.

**Safeguarding and recording (this lesson)**

- Tokens, counters and small things stay on the table: nothing goes in the mouth or ears. No matches, toothpicks or anything sharp. Keep small things away from younger brothers and sisters. Money is whole dollars only: no real money, notes or coins are needed.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper or a board up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. Mi's Challenge Quest stays flat on the table and is never held up or read out, and nothing is added up in front of the group. There is no private chat.
- Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. Counting aloud is quiet; a child may stay seated.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for 90 days after the program ends. The tutor then helps in the open session next term. The quest photos follow the same route and are opened only on the business WhatsApp, never on a personal phone. No private messages to children.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

## 2. Timing plan (90 minutes exactly)

| Minutes | Part | Min | Slides | Script fills |
|---|---|---|---|---|
| 0–2 | **Opening** | 2 | 1: Find One First | 1:33 |
| 2–8 | **Remember warm-up** | 6 | 3: Warm-up: share equally, Each and in all, A strip of equal parts | 4:08 |
| 8–12 | **Sneak-peek puzzles** | 4 | 1: Sneak-peek puzzles | 2:51 |
| 12–22 | **Explore** | 10 | 4: Explore: find one, Find many, Find one, then many, Find one, then many | 8:16 |
| 22–31 | **Mini-lesson** | 9 | 7: Find one first, Find one first, Write the two steps, Three words, A dozen, Your turn, Check: your turn | 7:29 |
| 31–35 | **Brain break** | 4 | 1: Move and count | 3:11 |
| 35–55 | **Mi's Challenge Quest** | 20 | 2: Mi's Challenge Quest, The Pearl Cave | 17:09 |
| 55–80 | **Workbook** | 25 | 5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page | 20:56 |
| 80–84 | **Talk time** | 4 | 2: Talk and explain, Back to the sneak-peek | 3:20 |
| 84–86 | **Look-back** | 2 | 1: Look-back | 1:38 |
| 86–90 | **Flex time and goodbye** | 4 | 1: Well done, inquirers! | 0:48 |
| | **Total** | **90** | | **71:21** |

The script needs 71:21 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 84:20. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 55 (start of workbook), minute 80 (talk time). If more than 2 minutes behind at minute 55, skip in this order: In a slow class the workbook part runs over: take the time from the 5-minute flex buffer and shorten choice time (pages 3 to 5 can go home); the optional slide and the optional exit ticket; the dozen slide in the Mini-lesson; never shorten Mi's Challenge Quest. Never skip the look-back or the camera and privacy moments.

## 3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

### 0:00 – 2:00 Opening

script 1:33 of 2:00

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| 0:00 | SLIDE | Show slide **Find One First** (s-title) |
| 0:15 | REC | *If you are watching the recording: hello! Get a pencil, paper, your workbook, tokens and your Quest Log. Pause or rewind any time.* |
| 0:26 | DO | Greet each child by first name as they arrive. Then start the recording. While it runs, avoid names. Check that you can see each child. [0:12] |
| 0:38 | SAY | Hello, inquirers! Please stay muted unless I invite you to speak. Cameras on if you can, so I can see you; cameras off is fine too. Pencil ready, and your tokens and your closed Quest Log on the table. |
| 0:58 | SAY | This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is unmuted. You may mute and keep your camera off. |
| 1:15 | SAY | Today we find one first, and then we find many. After our break we play Mi's Challenge Quest, a game with Mi, on our own paper. It is not a race, and nobody sees your answers. |
| | NOTE | Say the recording notice in the SAY line above in your own words after the recording has started: the recording has the tutor, the shared screen and any unmuted voice; children may mute and keep the camera off. The routine and the four inquirer rules were introduced earlier in the term: do not re-teach them. |
| | NOTE | Plain statement for families when it comes up: sharing within 20 and multiplying within 40 in whole dollars is Primary 1 work, but chaining the two steps with different numbers, a dozen, such and how many can be bought go beyond Primary 1 on purpose. |
| | NOTE | New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper to the camera. The first recording of the term shows the full routine. |

### 2:00 – 8:00 Remember warm-up

script 4:08 of 6:00

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| 2:00 | SLIDE | Show slide **Warm-up: share equally** (s-rem1) |
| 2:25 | REC | *Recording: join in! Say your answers out loud before I say them. Pause if you need longer.* |
| 2:34 | SAY | Twelve counters, three plates. Deal the counters one at a time, so every plate has the same. |
| 2:42 | ASK | How many counters are on each plate? [wait about 12 s for answers] |
| 2:58 | ASK | Six counters shared between two plates. How many on each plate? [wait about 12 s for answers] |
| 3:15 | ASK | Fifteen counters shared among three plates. How many on each plate? [wait about 12 s for answers] |
| 3:32 | SLIDE | Show slide **Each and in all** (s-rem2) |
| 3:58 | SAY | Three bags with four counters in each bag. Each bag has four. In all means all the bags together. |
| 4:07 | ASK | How many counters are there in all? [wait about 12 s for answers] |
| 4:22 | ASK | Five bags with two in each bag. How many in all? [wait about 12 s for answers] |
| 4:40 | ASK | Four groups of five? [wait about 12 s for answers] |
| 4:54 | SLIDE | Show slide **A strip of equal parts** (s-rem3) |
| 5:19 | SAY | A strip cut into four equal parts. The whole strip is twenty. |
| 5:25 | ASK | How big is each part? [wait about 12 s for answers] |

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| 5:40 | **ASK** | Three parts of five? [wait about 12 s for answers] |
| 5:54 | **ASK** | Six shared into two equal parts? [wait about 12 s for answers] |
| | **IF STUCK** | If a child is unsure about sharing, deal real counters one at a time onto plates or pieces of paper. For the bags, add one bag at a time and say the total each time. For the strip, count on in fives on fingers and count the parts. |
| | **NOTE** | Answers: 4, 3, 5; 12, 10, 20; 5, 15, 3. Children answer with a whisper, fingers or hands. Nobody writes. Do not say price or cost yet. Joining cold: all you need is counting to 20 and adding small numbers. |

## 8:00 – 12:00 Sneak-peek puzzles

script 2:51 of 4:00

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| 8:00 | **SLIDE** | Show slide **Sneak-peek puzzles** (s-sneak) |
| 8:25 | **REC** | *Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.* |
| 8:34 | **SAY** | Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is judged. |
| 8:47 | **READ** | Puzzle one. 4 pencils cost 8 dollars. How much does 1 pencil cost? You may use counters. |
| 8:56 | **CHILDREN** | Children share counters or draw, and write the number on their own paper. [0:40] |
| 9:36 | **READ** | Puzzle two. 3 apples cost 6 dollars. How much do 5 apples cost? |
| 9:43 | **CHILDREN** | Children write or draw their answer. [1:00] |
| 10:43 | **SAY** | Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time. |
| | **NOTE** | Puzzle 1 answer: $2. Puzzle 2 answer: $10 (1 apple costs $2, so 5 apples cost $10). Look for good ideas to praise later (a child who found 1 apple first). Do not write scores or keep a list. |

## 12:00 – 22:00 Explore

script 8:16 of 10:00

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| 12:00 | **SLIDE** | Show slide **Explore: find one** (s-explore1) |
| 12:25 | **REC** | *Recording: pause now. Use your tokens to find the cost of one, then of many. Press play when you have tried.* |
| 12:36 | **SAY** | Your hands are the teachers. Today a token is one dollar. Four pens cost twenty dollars. Count out twenty tokens, and put out four small things for the pens. |
| 12:50 | **CHILDREN** | Children count out 20 tokens and deal them one at a time onto 4 pens until every pen has the same. [1:30] |
| 14:20 | **ASK** | How much does 1 pen cost? [wait about 15 s for answers] |
| 14:38 | **SAY** | One pen costs five dollars. Twenty dollars shared equally among four pens is five dollars each. |
| 14:46 | **SLIDE** | Show slide **Find many** (s-explore2) |
| 15:11 | **SAY** | Round two. Now we know one pen costs three dollars. How much do four pens cost? Put three tokens by each pen. |
| 15:22 | **CHILDREN** | Children put 3 tokens by each of 4 pens and count all the tokens. [1:10] |
| 16:32 | **ASK** | How much do 4 pens cost? [wait about 15 s for answers] |
| 16:50 | **SAY** | Four groups of three dollars is twelve dollars. |
| 16:54 | **SLIDE** | Show slide **Find one, then many** (s-explore3) |
| 17:19 | **SAY** | Round three. Four pens cost twenty dollars. How much do three pens cost? First find one pen, then take three pens. |
| 17:30 | **CHILDREN** | Children share 20 tokens onto 4 pens, then take 3 pens and count their tokens. [1:40] |
| 19:10 | **ASK** | How much does 1 pen cost? [wait about 15 s for answers] |
| 19:28 | **ASK** | How much do 3 pens cost? [wait about 15 s for answers] |

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| 19:46 | SAY | Find one first, then find many. That is our idea for today. |
| | IF STUCK | A child who adds 20 and 4: ask, what are we sharing? A child who stops after finding 1: ask, how many pens do we want? A child with no tokens uses counters, beans or drawn circles. Money is only whole dollars; do not use coins or notes. |
| | NOTE | Main room only; children say their answers aloud and show fingers; you cannot see tokens on small tiles. Answers: 1 pen $5; 4 pens at $3 is $12; 1 pen $5 and 3 pens $15. A token stands for one dollar. Do not teach cents, coins or notes. |
| 19:52 | SLIDE | Show slide **Find one, then many** (s-explore3-ans) |

## 22:00 – 31:00 Mini-lesson

script 7:29 of 9:00

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| 22:00 | SLIDE | Show slide **Find one first** (s-m1) |
| 22:25 | REC | *Recording: keep watching and copy the examples. Rewind if you miss a step.* |
| 22:32 | SAY | Five cups cost fifteen dollars. How much do two cups cost? Step one: find one cup, by sharing the cost equally. Step two: find many, by counting that many ones. |
| 22:46 | ASK | How much does 1 cup cost? [wait about 15 s for answers] |
| 23:04 | ASK | How much do 2 cups cost? [wait about 15 s for answers] |
| 23:22 | SLIDE | Show slide **Find one first** (s-m1-ans) |
| 23:48 | SLIDE | Show slide **Write the two steps** (s-m2) |
| 24:12 | SAY | We write two steps. Step one: fifteen shared equally among five cups is three dollars. Step two: two groups of three dollars is six dollars. We check: five cups at three dollars each make fifteen dollars. |
| 24:30 | SLIDE | Show slide **Three words** (s-m3) |
| 24:56 | SAY | Three words. Each means for one. A dozen is twelve. Such means of the same kind: three such cups are three cups the same. |
| 25:08 | ASK | How many eggs are in a dozen eggs? [wait about 15 s for answers] |
| 25:26 | SLIDE | Show slide **A dozen** (s-m3b) |
| 25:52 | SAY | A dozen cookies is twelve cookies. A dozen is twelve, whatever there are twelve of. |
| 25:59 | SLIDE | Show slide **Your turn** (s-m4) |
| 26:24 | READ | Your turn. 6 pencils cost 18 dollars. How much does 1 pencil cost? How much do 4 pencils cost? |
| 26:34 | CHILDREN | Children work on their own paper, with tokens if they like. Wait 10 seconds before helping. [2:20] |
| 28:54 | SLIDE | Show slide **Check: your turn** (s-m4a) |
| 29:19 | READ | One pencil costs 3 dollars. Four pencils cost 12 dollars. Tick your own work with a tick. |
| | NOTE | Key words to use every time: find one first, find many, each, in all, cost. Step 1 shares, step 2 counts groups. The ÷ sign is not needed: say shared equally. Do not teach a rule to recite; the tokens and the strip are the method. Beyond Primary 1 on purpose: a dozen, such and the price of many from a different number are in Think harder. |
| | IF STUCK | If a child multiplies 15 and 5, ask: what do we know, the cost of 1 or the cost of 5? If a child stops after step 1, ask: how many do we want? If a child cannot share 18 among 6, deal 18 tokens onto 6 pencils one at a time. |

## 31:00 – 35:00 Brain break

script 3:11 of 4:00

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| 31:00 | SLIDE | Show slide **Move and count** (s-break) |
| 31:25 | REC | *Recording: stand up and join in! Count in twos and fives with the moves.* |
| 31:32 | SAY | Stand up and get a sip of water. We count in twos and stamp softly on each number. Then we count in fives and clap on each number. |

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| 31:46 | **CHILDREN** | Count in 2s to 20 with a soft stamp, then in 5s to 40 with a clap. Children may stay seated and tap the table. Then everyone takes out the closed Quest Log and a pencil. [2:25] |
| | **NOTE** | Quiet movement only. After the break say: please sit down, put the closed Quest Log flat on your table, and have a pencil ready. |

## 35:00 – 55:00 Mi's Challenge Quest

script 17:09 of 20:00

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| 35:00 | **SLIDE** | Show slide **Mi's Challenge Quest** (s-quest1) |
| 35:25 | **REC** | *Recording: Mi's Challenge Quest is not recorded. Skip ahead to the workbook. Play the quest as your tutor arranges.* |
| 35:34 | **SAY** | Now it is time for Mi's Challenge Quest! Mi the dolphin is swimming to the Pearl Cave, and eight gates are in the way. At each gate there is a puzzle for you. You work on your own paper with your quest partner beside you. |
| 35:57 | **SAY** | It is a game, not a race, and nobody gets a score. You may skip a gate and Mi will swim round it. I will not say any answers, and I will not show anyone's paper. |
| 36:15 | **DO** | Pause the recording now. Children open the closed Quest Log, with a pencil. Quest partners sit beside the child and may talk about how to start, but do not give answers. Paper flat on the table, never held up. Name on page 1. Counters and fingers are fine. [0:30] |
| 36:45 | **READ** | Page 1. Gate 1. Siti is 4th in a line. Ben stands right behind Siti. Part a: Ben is blank in the line. Part b: how many children are in front of Siti? Gate 2. Ravi has 26 marbles. Lily has 9 more marbles than Ravi. Part a: Lily has blank marbles. Part b: they have blank marbles altogether. Gate 3. Look at the ten-frames: 9 and 9. Part a: 9 plus 9 equals blank. Part b: write odd or even. 15 is blank. |
| 37:31 | **CHILDREN** | Children work quietly on page 1 (gates 1 to 3); partners may whisper about how, not about the answer. Look around the gallery; say nothing about right or wrong. If asked, re-read a gate to the whole group. [3:10] |
| 40:41 | **SAY** | Gates one, two and three are open! Mi swims on. Turn to page 2. |
| 40:48 | **READ** | Gate 4. Jenny has 3 jars. Each jar has 5 sweets. Part a: fill in the boxes: blank times blank equals blank. Part b: count in 2s: 2, 4, 6, blank, 10. Gate 5. Siti shares 16 stickers equally among 4 friends. Kumar puts 10 pens into boxes, 2 pens in each box. Part a: each of Siti's friends gets blank stickers. Part b: Kumar needs blank boxes. Gate 6. Mei has 5 bags. Each bag has 3 sweets. She gives 4 sweets to Ben. Part a: the bags have blank sweets. Part b: Mei has blank sweets left. |
| 41:42 | **CHILDREN** | Children work quietly on page 2 (gates 4 to 6). [3:30] |
| 45:12 | **SAY** | Gates four, five and six are open! |
| 45:15 | **MOVE** | One-minute stretch: stand, shake out your hands, sip some water. Paper stays on the table. |
| 45:55 | **SAY** | Turn to page 3. The Pearl Cave is just ahead. |
| 46:00 | **READ** | Gate 7. Each shape is a number. The heart plus the heart equals 18. Ben says the heart is 8. Part a: is Ben correct? Circle Yes or No. Part b: the heart is blank. Gate 8. 8 buns cost 16 dollars. Part a: 1 bun costs blank dollars. Part b: 5 buns cost blank dollars. The secret cave has one more puzzle, and it is optional. You may skip it. Ravi is 8th from the front and 8th from the back of a line. Mei says there are 16 children in the line. Is Mei correct? Circle Yes or No. Then write how many children are in the line. |
| 47:00 | **CHILDREN** | Children work quietly on page 3: gates 7 and 8, then the optional secret cave item if they wish. [3:50] |
| 50:50 | **SLIDE** | Show slide **The Pearl Cave** (s-quest2) |
| 51:15 | **SAY** | You reached the Pearl Cave! Make sure your name is on every page and close your Quest Log. Well done for trying your best. Grown-ups, please take one photo of each page today and send them to the business WhatsApp, as in the note. Mi's tutor will send you one line about what we saw, never a mark. |
| 51:44 | **DO** | Say nothing about answers. Remind the group that nothing is added up and no answer is read out. Restart the recording for the workbook. Children who did not finish may use a few minutes of the workbook time. [0:25] |

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| | **IF STUCK** | A child who says they cannot do a gate: say to the whole group that skipping is fine, Mi swims round it, and they may come back. A child who is upset: the grown-up may stop at any time; do not single the child out. Never give a hint to one child about an answer. |
| | **NOTE** | The answers, your private notes sheet for each child, the group advice sheet and the one-line templates for the grown-up are in L20_Quest_Tutor_Notes.pdf. Do not collect or look at any paper during the lesson. Photos arrive only on the program's business WhatsApp and the line goes to the parent only; never a message to a child, and never a personal phone. The exit ticket on workbook page 5 is for home this lesson: the quest is the exit. |

## 55:00 – 80:00 Workbook

script 20:56 of 25:00

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| 55:00 | **SLIDE** | Show slide **Workbook page 1: Start here** (s-ws1) |
| 55:25 | **REC** | *Recording: pause now and do workbook page 1. When you finish a page, press play to check it.* |
| 55:34 | **READ** | Page 1, Start here. Question 1: 4 pens cost 16 dollars. How much does 1 pen cost? Question 2: 1 pen costs 2 dollars. How much do 7 pens cost? Fill in the blanks. Question 3: 6 buns cost 12 dollars. How much does 1 bun cost? Look at the strip. |
| 56:02 | **CHILDREN** | Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [7:00] |
| 63:02 | **SLIDE** | Show slide **Check page 1** (s-ws1ans) |
| 63:27 | **READ** | Question 1: 4 dollars. Question 2: 14 dollars. Question 3: 2 dollars. Tick your own page with a tick. |
| 63:37 | **SAY** | After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time. |
| 63:46 | **SLIDE** | Show slide **Workbook page 2: Start here** (s-ws2) |
| 64:11 | **READ** | Page 2. Question 4: 3 books cost 15 dollars. Fill in the blanks. Look at the strip. Question 5: 5 pens cost 15 dollars. Lily says 1 pen costs 4 dollars. Is Lily correct? Question 6: each book costs 4 dollars. Kumar buys 3 books. How much do the 3 books cost? |
| 64:40 | **CHILDREN** | Page 2, questions 4 to 6. [8:00] |
| 72:40 | **SLIDE** | Show slide **Check page 2** (s-ws2ans) |
| 73:05 | **READ** | Question 4: 5 dollars and 10 dollars. Question 5: no, 1 pen costs 3 dollars. Question 6: 12 dollars. |
| 73:15 | **SLIDE** | Show slide **Choose your next page** (s-choice) |
| 73:40 | **SAY** | When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too. |
| 73:56 | **MOVE** | Stand and stretch for 30 seconds, then sit down and choose. |
| 74:26 | **CHILDREN** | Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [1:30] |
| | **IF STUCK** | Q1: share 16 between 4 pens. Q2: put 2 by each of 7 pens. Q3: count the parts of the strip, then share 12. Q4: find 1 book first, then 2 books. Q5: put Lily's number into the story: 5 groups of 4 is 20. Q6: 3 groups of 4. |
| | **IF STUCK** | A child who is still stuck ticks I am still stuck on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp (never a personal phone). You then help in the open session at the start of next term. Never a private message to a child. |
| | **NOTE** | Think harder chains the two steps with different numbers and uses a dozen, such and how many can be bought (beyond Primary 1 on purpose). Go further has two kinds of item, money left over and a comparison of two shops. Do not announce who is on which page. The check came first, so some children are tired, and Start here takes a middle child about 16 minutes: say that pages 3 to 5 are for home. Slow class: take the extra minutes from the flex time and do not skip the answer slides. |

## 80:00 – 84:00 Talk time

script 3:20 of 4:00

| | | |
|---|---|---|
| 80:00 | SLIDE | Show slide **Talk and explain** (s-talk) |
| 80:25 | REC | *Recording: pause. Tell a grown-up if Aisha is right, and test it with tokens.* |
| 80:32 | READ | Aisha says: 3 books cost 12 dollars, so 6 books cost 24 dollars. Is Aisha right? How could we check? |
| 80:43 | CHILDREN | Test Aisha with tokens: 12 on 3 books, then 6 books. [0:35] |
| 81:18 | CHILDREN | Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:20] |
| 81:38 | ASK | Who would like to share? Volunteers first. A pass is always fine. [wait about 35 s for answers] |
| 82:19 | SLIDE | Show slide **Back to the sneak-peek** (s-sneakans) |
| 82:44 | SAY | Remember the sneak-peek? Four pencils cost eight dollars, so one pencil costs two dollars. Three apples cost six dollars, so one apple costs two dollars, and five apples cost ten dollars. |
| 82:59 | ASK | Did anyone change their mind while we worked? You can just nod. [wait about 15 s for answers] |
| | NOTE | Model answer: yes. 1 book costs $4, so 6 books cost 6 groups of $4, which is $24. Or 6 books are 2 lots of 3 books, so 2 lots of $12. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed. |

## 84:00 – 86:00 Look-back

script 1:38 of 2:00

| | | |
|---|---|---|
| 84:00 | SLIDE | Show slide **Look-back** (s-look) |
| 84:25 | REC | *Recording: pause and fill in your look-back box. You can pass.* |
| 84:30 | SAY | Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick I am still stuck, ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you. |
| 85:03 | CHILDREN | Children fill in the look-back box on page 5. [0:35] |

## 86:00 – 90:00 Flex time and goodbye

script 0:48 of 4:00

| | | |
|---|---|---|
| 86:00 | SLIDE | Show slide **Well done, inquirers!** (s-end) |
| 86:25 | SAY | Well done, inquirers, for the whole term. Next term starts with What Time Is It?, the first lesson of Term 3. You need your Term 3 workbook, a pencil and paper. Your tutor will tell you the start date. Goodbye, and thank you to the grown-ups. |
| | SLIDE | Show slide **Go further: do we always find one first?** (s-opt) *(optional)* |
| | SAY | Optional, only if you are on time: 2 pens cost 6 dollars. How much do 4 pens cost? Do we need to find 1 pen first? |
| | CHILDREN | Short think time, then take two answers. [1:30] |
| | SLIDE | Show slide **If we have time: pick one** (s-menu) *(optional)* |
| | SLIDE | Show slide **Exit ticket** (s-exit) *(optional)* |
| | READ | Optional exit ticket. Question 1: 3 pencils cost 9 dollars. How much does 1 pencil cost? Question 2: each sticker costs 5 dollars. How much do 3 stickers cost? Option 1: 8 dollars. Option 2: 15 dollars. Option 3: 2 dollars. Option 4: 53 dollars. Write its number in the bracket. Question 3: 5 cakes cost 10 dollars. Fill in the blanks: 1 cake costs blank dollars; 4 cakes cost blank dollars. |
| | CHILDREN | Children answer E1 to E3 on page 5 on their own. [1:40] |
| | SLIDE | Show slide **Check the exit ticket** (s-exitans) *(optional)* |
| | READ | Question 1: 3 dollars. Question 2: option 2, 15 dollars. Question 3: 2 dollars and 8 dollars. Tick your own ticket with a tick. |

**SAY** Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).

**NOTE** This is the standing buffer. A child who needs more time for the quest may use it. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 4 pens are 2 lots of 2 pens, so $12; finding one first always works and doubling is a quick way here. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

## 3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

| Activity | Minutes | You say | Recording cue | Answer |
|---|---|---|---|---|
| **Does 1 cost more if we buy more?** | 4 (optional) | Mei says: 2 pens cost $6, so 1 pen costs $3. At 4 pens for $12, 1 pen costs $6. Is Mei right? Test it with tokens. | *Recording: pause. Mei says 1 pen costs $3 for 2 pens at $6, but $6 for 4 pens at $12. Is she right?* | No. $12 shared equally among 4 pens is $3 each, the same as before. 1 pen costs the same when each pen costs the same, so we say such pens or each. |
| **Make a shop puzzle for your grown-up** | 4 (optional) | Choose a price for 1 pen and a number of pens. Write how much they cost in all. Ask your grown-up to find the price of 1 pen, then check with tokens. | *Recording: pause. Choose a price for 1 pen. Write the total for some pens. Ask a grown-up to find the price of 1.* | No single answer. Check by sharing: with 4 pens at $5 each the total is $20, and $20 shared among 4 pens is $5. |
| **Teach it back** | 3 (optional) | Pretend your grown-up or a toy has never found the price of one. Use tokens to show how 3 books cost $12, so 1 book costs $4. | *Recording: pause. Show a grown-up how to share 12 tokens among 3 books to find the cost of 1 book.* | 12 tokens shared equally among 3 books is 4 tokens each, so 1 book costs $4. |

# 4. Questions, answers, hints and common mistakes

## Sneak-peek puzzles (minutes 8 to 12)

**Puzzle 1 (access):** 4 pencils cost $8. How much does 1 pencil cost?

**Answer: $2**

**Puzzle 2 (think):** 3 apples cost $6. How much do 5 apples cost?

**Answer: $10. 3 apples cost $6, so 1 apple costs $2. 5 apples cost 5 groups of $2, which is $10.**

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

## Start here (workbook pages 1 and 2)

**S1** **4 pens cost $16. How much does 1 pen cost? Use tokens or draw.**



1 pen costs $☐.

**Answer: $4**

**Hint (say this first):** Share $16 equally among 4 pens. Use tokens.
**Worked solution:** $16 shared equally among 4 pens is $4 each. Check: 4 groups of $4 make $16.
**Common mistakes:** Writing 20 (adding 16 and 4).; Writing 64 (multiplying).; Writing 16 (the total).
Time for a middle child: about 2.5 min.

**S2** **1 pen costs $2. How much do 7 pens cost? Fill in the blanks.**



(a) 7 × 2 = ☐☐
(b) 7 pens cost $☐☐.

**Answer: 14; $14**

**Hint (say this first):** Put 2 tokens by each pen and count them all.
**Worked solution:** 7 groups of $2 is 2 + 2 + 2 + 2 + 2 + 2 + 2 = 14, so 7 pens cost $14.
**Common mistakes:** Writing 9 (adding 2 and 7).; Writing 2 (the cost of 1 pen).
Time for a middle child: about 2.5 min.

**S3** **6 buns cost $12. How much does 1 bun cost? Look at the strip.**




1 bun costs $☐.

**Answer: $2**

**Hint (say this first):** The strip has 6 equal parts. Share $12 into 6 parts.
**Worked solution:** $12 shared equally among 6 buns is $2 each. Check: 6 groups of $2 make $12.
**Common mistakes:** Writing 6 (the number of buns).; Writing 72 (multiplying).
Time for a middle child: about 2.5 min.

**S4** **3 books cost $15. Fill in the blanks. Look at the strip.**




(a) 1 book costs $☐.
(b) 2 books cost $☐☐.

**Answer: $5; $10**

**Hint (say this first):** Find 1 book first. Then find 2 books.
**Worked solution:** $15 shared equally among 3 books is $5 each. 2 books cost 2 groups of $5, which is $10. Check: 3 groups of $5 make $15.
**Common mistakes:** Writing 15 for 2 books.; Writing 7 (adding 5 and 2).; Stopping after the first part.
Time for a middle child: about 3 min.

**S5** **5 pens cost $15. Lily says 1 pen costs $4. Look at the picture.**



(a) Is Lily correct? Circle Yes or No.
(b) 1 pen costs $☐.

**Answer: No; $3**

**Hint (say this first):** Put Lily's number back: 5 groups of $4.
**Worked solution:** 5 groups of $4 make $20, not $15. So Lily is not correct. $15 shared equally among 5 pens is $3 each. Check: 5 groups of $3 make $15.
**Common mistakes:** Saying Yes because 4 is close to 3.; Writing 15 for 1 pen.
Time for a middle child: about 3 min.

**S6** **Each book costs $4. Kumar buys 3 books. How much do the 3 books cost? Use tokens or draw.**



The 3 books cost $☐☐.

**Answer: $12**

**Hint (say this first):** Each means for one. Count 3 groups of $4.
**Worked solution:** 3 groups of $4: 4 + 4 + 4 = 12, so the 3 books cost $12.
**Common mistakes:** Writing 7 (adding 4 and 3).; Writing 4.
Time for a middle child: about 2.5 min.

## Think harder (page 3): the taught idea in a new setting

**T1** **4 cups cost $12. Fill in the blanks. Look at the strip.**




(a) 1 cup costs $☐.
(b) 7 cups cost $☐☐.

**Answer: $3; $21**

**Hint (say this first):** Find 1 cup first. Then find 7 cups.
**Worked solution:** $12 shared equally among 4 cups is $3 each. 7 cups cost 7 groups of $3, which is $21. Check: 4 groups of $3 make $12.
**Common mistakes:** Writing 84 (12 × 7).; Writing 19 (12 + 7).; Stopping after the first part.
Time for a middle child: about 4 min.

**T2** **A dozen buns cost $24. Fill in the blanks.**

(a) 1 bun costs $☐.
(b) 3 buns cost $☐.

**Answer: $2; $6**

**Hint (say this first):** A dozen is 12. Share $24 equally among 12 buns.
**Worked solution:** A dozen is 12, so 12 buns cost $24. $24 shared equally among 12 buns is $2 each. 3 buns cost 3 groups of $2, which is $6.
**Common mistakes:** Using 24 shared among 2 buns.; Writing 24 for 3 buns.
Time for a middle child: about 3.5 min.

**T3** **Wei pays $24 for 6 such cups. Fill in the blanks.**

(a) 1 cup costs $☐.
(b) 4 such cups cost $☐☐.

**Answer: $4; $16**

**Hint (say this first):** Such means the cups are the same, so each costs the same.
**Worked solution:** $24 shared equally among 6 cups is $4 each. 4 such cups cost 4 groups of $4, which is $16.
**Common mistakes:** Writing 24 for 4 cups.; Writing 8 (4 + 4).
Time for a middle child: about 3.5 min.

**T4** **4 pens cost $8. Ben has $10. Fill in the blanks.**

(a) 1 pen costs $☐.
(b) Ben can buy ☐ pens.

**Answer: $2; 5 pens**

**Hint (say this first):** Find 1 pen first. Then count how many groups of $2 fit in $10.
**Worked solution:** $8 shared equally among 4 pens is $2 each. $10 holds 5 groups of $2, so Ben can buy 5 pens.
**Common mistakes:** Writing 4 pens (the number in the story).; Writing 20 (10 × 2).
Time for a middle child: about 4 min.

## Go further (page 4): mastery and extension

**G1** **2 hats cost $10. 3 scarves cost $18. Jenny buys 1 hat and 2 scarves. Fill in the blanks.**

(a) 1 hat costs $☐.
(b) 1 scarf costs $☐.
(c) Jenny pays $☐☐.

**Answer: $5; $6; $17**

**Hint (say this first):** Find 1 hat and 1 scarf first.
**Worked solution:** $10 shared equally among 2 hats is $5 each. $18 shared equally among 3 scarves is $6 each. 2 scarves cost $12. 1 hat and 2 scarves cost $5 + $12 = $17.
**Common mistakes:** Adding 10 and 18.; Counting 1 scarf instead of 2.
Time for a middle child: about 5 min.

**G2** **8 pens cost $24. Mei has $30. She buys 5 pens. Fill in the blanks.**

(a) 1 pen costs $☐.
(b) 5 pens cost $☐☐.
(c) Mei has $☐☐ left.

**Answer: $3; $15; $15**

**Hint (say this first):** Find 1 pen first, then 4 pens, then what is left from $20.
**Worked solution:** $24 shared equally among 8 pens is $3 each. 5 pens cost $15. $30 - $15 = $15 is left.
**Common mistakes:** Taking 24 away from 30.; Forgetting to take the cost away from $30.
Time for a middle child: about 5 min.

**G3** **Shop A sells 4 pens for $12. Shop B sells 5 pens for $20. Which shop sells 1 pen for less?**

| Shop | Pens | Cost in $ |
|---|---|---|
| A | //// | 12 |
| B | ///// | 20 |

(a) At Shop A, 1 pen costs $☐.
(b) At Shop B, 1 pen costs $☐.
(c) Circle Shop A or Shop B.

**Answer: $3; $4; Shop A**

**Hint (say this first):** Find the cost of 1 pen in each shop first.
**Worked solution:** Shop A: $12 shared among 4 pens is $3. Shop B: $20 shared among 5 pens is $4. $3 is less than $4, so Shop A sells 1 pen for less.
**Common mistakes:** Choosing Shop B because 20 is more.; Comparing $12 and $20 without finding 1 pen.
Time for a middle child: about 5 min.

## Investigation (page 5): Which pack costs less?

**Easy first step:** A pack of 2 pens costs $8. How much does 1 pen cost?
**Then:** A pack of 5 pens costs $20. Does 1 pen cost less in this pack? Why?

**Model answer: No. In both packs 1 pen costs $4 ($8 shared between 2 pens, and $20 shared among 5 pens), so 1 pen does not cost less in the bigger pack. Finding the price of one lets us compare packs of different sizes.**

**Look for:** A child who finds $3 for both packs, or who says 1 pen costs the same in both. Credit any clear reason. A child who says the bigger pack costs more is right about the total but has not compared one pen.
**Common mistakes:** Saying 1 pen costs less in the pack of 5 because it has more pens.; Comparing $6 and $15 without finding 1 pen.

### Talk and explain (minutes 80 to 84)

**Aisha says: 3 books cost $12, so 6 books cost $24. Is Aisha right? How could we check?** Sentence starter: I think … because …

> **Model answer: Yes. Find 1 book first: $12 shared equally among 3 books is $4. 6 books cost 6 groups of $4, which is $24. We can also see that 6 books are 2 lots of 3 books, and 2 lots of $12 is $24. We can check with tokens.**

**Look for:** A child who finds the cost of 1 book first, or who sees that 6 books are 2 lots of 3 books. Not just yes or no. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

### Exit ticket (minutes 35 to 55): 3 questions

**E1** **3 pencils cost $9. How much does 1 pencil cost?**

1 pencil costs $☐.

> **Answer: $3**

**Worked solution:** $9 shared equally among 3 pencils is $3 each.
**Common mistakes:** Writing 12 (adding 9 and 3).; Writing 27 (multiplying).

**E2** **Each sticker costs $5. How much do 3 stickers cost? Write its number in the bracket.**

> **Answer: (2) $15**

**Worked solution:** Each sticker costs $5, so 3 stickers cost 3 groups of $5, which is $15.
**Common mistakes:** Choosing (1) $8 by adding 5 and 3.; Choosing (3) $2 by taking 3 from 5.

**E3** **5 cakes cost $10. Fill in the blanks.**

(a) 1 cake costs $☐.
(b) 4 cakes cost $☐.

> **Answer: $2; $8**

**Worked solution:** $10 shared equally among 5 cakes is $2 each. 4 cakes cost 4 groups of $2, which is $8.
**Common mistakes:** Writing 10 for 4 cakes.; Writing 9 (5 + 4).

### Look-back (minutes 84 to 86)

Children tick or write: *Something I did well*; *The hardest part was* (sharing to find one; finding many from one; using the strip; something else); *What helped me* (tokens; drawing a strip; checking my answer; I was not stuck); *Next time I will try* (using tokens; drawing a strip; checking my answer; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

## 5. Home practice and online practice

| # | Tier | Question | Answer |
|---|---|---|---|
| H1 | Start here | 5 pens cost $10. How much does 1 pen cost? $___ | $2 |
| H2 | Start here | Each cup costs $3. Fill in the blanks. 6 × 3 = ___ . 6 cups cost $___ (× means groups of) | 18; $18 |
| H3 | Start here | 4 rubbers cost $12. How much do 2 rubbers cost? $___ | $6 |
| H4 | Think harder | A dozen eggs cost $36. How much do 4 eggs cost? $___ | $12 |
| H5 | Go further | 3 mugs cost $12. 2 plates cost $10. Siti buys 1 mug and 1 plate. How much does Siti pay? $___ | $9 |

Online practice (`out/l20/practice`, key `t2l20`): 12 items; 10 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no

scores, no streaks.

## 6. Parent line for the term report

Your child is learning to find the cost of one by sharing equally, and then the cost of many. This week: pens, buns and books in whole dollars.

Term line (plan, section 5): Your child is learning to draw a story, to use groups and sharing, and to test a claim with an example. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

## 7. Slide list

| # | Time | Slide | On screen | Speaker note |
|---|---|---|---|---|
| 1 | 0–2 | **Find One First**<br>s-title | TODAY WE WILL: / Find the cost of one, then of many. / Play Mi's Challenge Quest. | Lesson 10 of Term 2, 90 minutes, the last lesson of the term. Objective: find the cost of one by sharing a total equally, then the cost of many; use each, a dozen and such. This lesson also holds Mi's Challenge Quest (20 minutes, after the brain break, a game with Mi on private paper). Plain statement for families: dividing within 20 and multiplying within 40 in whole dollars is Primary 1 work; finding the price of 7 from the price of 3, a dozen, such and how-many-can-be-bought go past the school Primary 1 syllabus on purpose. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 22, mini-lesson 22 to 31, brain break 31 to 35, Mi's Challenge Quest 35 to 55, workbook 55 to 80, talk 80 to 84, look-back 84 to 86, flex time and goodbye 86 to 90 (the buffer: do not start anything new). Say the recording notice in one line: you and the screen only; a child may keep the camera off. |
| 2 | 2–8 | **Warm-up: share equally**<br>s-rem1 | Deal the counters one at a time. Same number on each plate. / [picture: groups] / [picture: table] | Twelve counters are shared equally onto 3 plates. Children deal counters one at a time (or imagine it). Ask: how many on each plate (4); 6 counters shared between 2 plates (3); 15 counters shared among 3 plates (5). Children answer aloud or with fingers. No writing, no marks. A child who is unsure deals real counters or small things onto three plates or three pieces of paper. |
| 3 | 2–8 | **Each and in all**<br>s-rem2 | [picture: groups] / [picture: table] / Each bag has the same number. | Three bags with 4 counters in each bag. We say each bag has 4, and in all means every bag together. Ask: how many counters in all (12); 5 bags with 2 in each bag (10); 4 groups of 5 (20). This is Term 2 Lessons 5 and 8 taught again. A child who is unsure adds one bag at a time and says the total each time. |
| 4 | 2–8 | **A strip of equal parts**<br>s-rem3 | [picture: bar_model] / 4 equal parts. The whole strip is 20. | A strip is cut into 4 equal parts, and the whole strip is 20. Ask: how big is each part (5); 3 parts of 5 (15); 6 shared into 2 equal parts (3). This is Term 2 Lesson 8 taught again: a strip shows equal groups. A child who is unsure counts 5, 10, 15, 20 on fingers and counts the fingers. |
| 5 | 8–12 | **Sneak-peek puzzles**<br>s-sneak | Ideas we have not learnt yet. A guess is a great start. Nobody is judged. / PUZZLE 1: anyone can try: / 4 pencils cost $8. / How much does 1 pencil cost? / You may use counters. / PUZZLE 2: think: / 3 apples cost $6. / How much do 5 apples cost? | Puzzle 1 (4 pencils for $8; 1 pencil is $2) is an access question: any child can share 8 counters onto 4 pencils. Puzzle 2 is a thinking question on the lesson idea: find one first. 1 apple costs $2 and 5 apples cost $10. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Nobody holds paper to the camera. There is no private chat. Answers: $2; $10. Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list. |
| 6 | 12–22 | **Explore: find one**<br>s-explore1 | [picture: table] / 4 pens cost $20. Count out 20 tokens. / Share the tokens equally onto 4 pens. / How much does 1 pen cost? | Round 1, about 4 minutes. Children use their $1 tokens (the kit page or counters) and 4 small things as the pens. 4 pens cost $20: they count out 20 tokens and deal them one at a time onto the 4 pens until every pen has the same. Ask: how much does 1 pen cost (5). Say: 1 pen costs $5, because $20 shared equally among 4 pens is $5 each. Money is whole dollars only, and a token stands for $1. If a family has no shop items, 4 pieces of paper are fine. Answer: $5. |
| 7 | 12–22 | **Find many**<br>s-explore2 | [picture: table] / 1 pen costs $3. Put 3 tokens by each pen. / Do this for 4 pens. / Count all the tokens. | Round 2, about 3.5 minutes. Now we know the cost of one: 1 pen costs $3. For 4 pens children put 3 tokens by each pen and count all the tokens (12). Ask: how much do 4 pens cost (12). Say: 4 groups of $3 is $12. Link to Term 2 Lesson 5: groups of. Answer: $12. A child who loses count puts the tokens in a row of 3 and counts in 3s. |

| | | | | |
|---|---|---|---|---|
| 8 | 12–22 | **Find one, then many**<br>s-explore3 | [picture: bar_model] / 4 pens cost $20. / How much do 3 pens cost? | Round 3, about 3.5 minutes. 4 pens cost $20. How much do 3 pens cost? Children share 20 tokens onto 4 pens (5 each) and then take 3 pens (15). The strip shows 4 equal parts for the 4 pens, and 3 parts are the 3 pens. Ask: how much does 1 pen cost (5); how much do 3 pens cost (15). Say: find one first, then find many. Answers: $5 and $15. The steps (find 1 pen first, then take 3 parts) appear on the next slide, after the children have tried. |
| 9 | 12–22 | **Find one, then many**<br>s-explore3-ans | [picture: bar_model] / 4 pens cost $20. / How much do 3 pens cost? / Find 1 pen first: share $20 into 4 equal parts. / Then take 3 parts. | Round 3, about 3.5 minutes. 4 pens cost $20. How much do 3 pens cost? Children share 20 tokens onto 4 pens (5 each) and then take 3 pens (15). The strip shows 4 equal parts for the 4 pens, and 3 parts are the 3 pens. Ask: how much does 1 pen cost (5); how much do 3 pens cost (15). Say: find one first, then find many. Answers: $5 and $15. The steps (find 1 pen first, then take 3 parts) appear on the next slide, after the children have tried. (answer shown after the children have answered) |
| 10 | 22–31 | **Find one first**<br>s-m1 | [picture: bar_model] | 5 cups cost $15. How much do 2 cups cost? Step 1 finds 1 cup: $15 shared equally among 5 cups. Step 2 finds 2 cups: 2 groups of that. Ask: how much does 1 cup cost (3); how much do 2 cups cost (6). The steps card appears on the next slide, after the children have answered. Children say each number first. Always say find one first. |
| 11 | 22–31 | **Find one first**<br>s-m1-ans | [picture: bar_model] / STEPS: / Step 1: find 1. Share the cost equally. / Step 2: find many. Count that many ones. | 5 cups cost $15. How much do 2 cups cost? Step 1 finds 1 cup: $15 shared equally among 5 cups. Step 2 finds 2 cups: 2 groups of that. Ask: how much does 1 cup cost (3); how much do 2 cups cost (6). The steps card appears on the next slide, after the children have answered. Children say each number first. Always say find one first. (answer shown after the children have answered) |
| 12 | 22–31 | **Write the two steps**<br>s-m2 | [picture: table] / Check: 5 cups at $3 each make $15. | Write the two steps: Step 1 is 15 shared equally among 5, which is 3. Step 2 is 2 groups of 3, which is 6. Check by putting the cost of one back: 5 groups of $3 is $15. Say each step as a short sentence. Either reading order of a groups line is accepted in a child answer. Answers: $3 and $6. |
| 13 | 22–31 | **Three words**<br>s-m3 | Word \| What it means \| Example / each \| for one \| $2 each: 1 pen costs $2 / a dozen \| 12 \| a dozen eggs: 12 eggs / such \| of the same kind \| 3 such cups: 3 cups the same | Three words for word problems. Each means for one. A dozen is 12. Such means of the same kind, so 6 such cups are 6 cups the same, with the same price. Ask: what is a dozen (12); a dozen eggs is how many eggs (12); if 3 such cups cost $6, does each cup cost the same (yes). A dozen and such are used in Think harder. |
| 14 | 22–31 | **A dozen**<br>s-m3b | [picture: objects] / A dozen cookies is 12 cookies. | Count the cookies together: 12. A dozen is 12, whatever it is 12 of. A dozen eggs, a dozen pens. Ask: how many pens are in a dozen (12). This slide is optional to shorten: cut it first if the Mini-lesson runs over. |
| 15 | 22–31 | **Your turn**<br>s-m4 | [picture: bar_model] / How much does 1 pencil cost? How much do 4 pencils cost? | Each child works on their own paper with tokens if they like. 6 pencils cost $18. Step 1: 1 pencil costs $3. Step 2: 4 pencils cost 4 groups of $3, which is $12. Walk the cameras; do not ask children to hold papers up. Wait 10 seconds before helping. Answers: $3 and $12. |
| 16 | 22–31 | **Check: your turn**<br>s-m4a | Step \| What we do \| Cost / Step 1 \| 1 pencil: $18 shared among 6 pencils \| $3 / Step 2 \| 4 pencils: 4 groups of $3 \| $12 | Answers: 1 pencil costs $3 and 4 pencils cost $12. Read each answer aloud so a child watching the recording can mark their own work. Children tick their own work with a tick. Never ask a child to show a wrong answer to the group. |
| 17 | 31–35 | **Move and count**<br>s-break | Count in twos. Stamp for each one. / Count in fives. Clap for each one. / Stand up. Sip some water. | Four minutes. Stand up on camera. Count in twos to 20 and stamp softly on each number, then count in fives to 40 and clap on each number. No equipment, no running, a little clear space. A child may stay seated and tap the table. Do not give prizes or compare children. Then ask everyone to take out the closed Quest Log and a pencil, ready for Mi's Challenge Quest. |

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| 18 | 35–55 | **Mi's Challenge Quest**<br>s-quest1 | Your own paper. Mi swims with you. / Open your Quest Log. Name on page 1. / Pencil ready. Counters are fine. / Your quest partner sits beside you. / I read each gate. Then you write. / You may skip a gate. It is not a race. / AFTER: / A grown-up takes a photo of each page and sends it to the program's business WhatsApp. / Nothing is shown to the group. | Introduce the quest for about a minute: Mi the dolphin swims to the Pearl Cave through 8 gates and an optional secret cave (about 20 minutes), on your own private paper with a quest partner beside you; a game, not a race; you may skip any gate; no answers are read out; nothing is shown to the group; no clock or timer is shown. Pause the recording first. Children take out the closed Quest Log only now. The paper lies flat on the table. Quest partners sit beside the child and may talk about how, but do not give answers. Follow the quest tutor notes (L20_Quest_Tutor_Notes.pdf). The number is in the grown-up note, never on a slide. |
| 19 | 35–55 | **The Pearl Cave**<br>s-quest2 | Make sure your name is on every page. / Close the Quest Log. Put your pencil down. / A grown-up takes one photo of each page, flat and from above. / A grown-up sends the photos to the program's business WhatsApp number in the grown-up note, and nowhere else. / WELL DONE: / You reached the Pearl Cave! Mi will tell your grown-up one thing we saw. | Say thank you. The grown-up takes one clear photo of each of the 3 pages, with the whole page in view and the name on each page, and sends them today to the program's business WhatsApp, as in the grown-up note. The tutor never asks a child for the sheet and never comments on a child's work in front of the group. Remind the group that no answers will be read out and nothing is added up. Children who did not finish may use a few minutes of the workbook time to finish. Restart the recording for the workbook. The number is in the grown-up note, never on a slide. |
| 20 | 55–80 | **Workbook page 1: Start here**<br>s-ws1 | Questions 1 to 3. Take your time. / questions S1, S2, S3 | Read questions 1 to 3 aloud. Children work on their own with tokens; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder. Question 1: 4 pens cost $16. Question 2: 1 pen costs $2 and we want 7 pens. Question 3: a strip of 6 buns for $12. |
| 21 | 55–80 | **Check page 1**<br>s-ws1ans | answers S1, S2, S3 | Read each answer aloud. Children tick their own page with a tick. Never ask a child to show a wrong answer to the group. Answers: Q1 $4; Q2 $14; Q3 $2. |
| 22 | 55–80 | **Workbook page 2: Start here**<br>s-ws2 | Questions 4 to 6. Take your time. / questions S4, S5, S6 | Read questions 4 to 6 aloud. Question 4: 3 books cost $15; find 1 book, then 2 books. Question 5: 5 pens cost $15 and Lily says 1 pen costs $4; is she correct? Question 6: each book costs $4 and Kumar buys 3 books. |
| 23 | 55–80 | **Check page 2**<br>s-ws2ans | answers S4, S5, S6 | Read each answer aloud so a child watching the recording can mark their own work. Never ask a child to show a wrong answer to the group. Answers: Q4 $5 and $10; Q5 No, 1 pen costs $3; Q6 $12. |
| 24 | 55–80 | **Choose your next page**<br>s-choice | Page \| What it is / Think harder (page 3) \| 7 from 3, a dozen, such / Go further (page 4) \| Two kinds, and which shop? / Investigation (page 5) \| Which pack costs less? / Check \| Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too. | Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes. |

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| 25 | 80–84 | **Talk and explain**<br>s-talk | AISHA SAYS: / 3 books cost $12. / So 6 books cost $24. / [picture: bar_model] / Is Aisha right? How could we check? / I think ... because ... | One minute of talk to the grown-up next to you, or to your screen. Children test Aisha with tokens: 12 tokens on 3 books is 4 each, and 6 books at 4 each is 24; or 6 books are 2 lots of 3 books, so 2 lots of $12. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: yes. 1 book costs $4, so 6 books cost 6 groups of $4, which is $24. Repeat each child answer in your own words for the recording. |
| 26 | 80–84 | **Back to the sneak-peek**<br>s-sneakans | [picture: table] / [picture: table] / 1 pencil costs $2. 5 apples cost $10. | Reveal both puzzles now. Say: many of you wrote a good reason. Do not read out anyone earlier answer. 4 pencils cost $8, so 1 pencil costs $2. 3 apples cost $6, so 1 apple costs $2, and 5 apples cost 5 groups of $2, which is $10. It was the same idea: find one first. |
| 27 | 84–86 | **Look-back**<br>s-look | [look-back box] | Two minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never graded, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide); open next term with one catch-up question for the whole group; never a direct message. |
| 28 | 86–90 | **Well done, inquirers!**<br>s-end | Show a grown-up one thing you found out this term. / NEXT TERM: / Term 3: What Time Is It? (the first lesson) / Have ready: the first Term 3 workbook, a pencil and paper. Your tutor will tell you the start date. | Say goodbye and thank the grown-ups. Minutes 86 to 90 are flex time: the buffer. A child who needs longer to finish the quest may use it; otherwise finish any open item. Do not start anything new. If the whole class is ahead, show the optional slides and use the Spare-time menu from the tutor guide. Remind families that the tutor sends one line about what we saw in Mi's Challenge Quest, never a mark. |
| 29 | optional | **Go further: do we always find one first?**<br>s-opt | [picture: bar_model] / 2 pens cost $6. How much do 4 pens cost? / Do we need to find 1 pen first? | Optional, only if you are on time. 2 pens cost $6. 4 pens are 2 lots of 2 pens, so 4 pens cost 2 lots of $6, which is $12. We do not need to find 1 pen first when the new number is a whole number of lots. Finding one first always works; doubling is a quick way here. Take two answers and move on. |
| 30 | optional | **If we have time: pick one**<br>s-menu | Activity \| What we do / Does 1 cost more? \| Mei says 1 pen costs $6 at 4 pens for $12. Is she right? / Make a shop puzzle \| Write a price puzzle for your grown-up / Teach it back \| Show a toy how to find 1 first / Optional. Everyone may join. | Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum. |
| 31 | optional | **Exit ticket**<br>s-exit | Optional. On your own. No looking back. / questions E1, E2, E3 | OPTIONAL, only if the whole class has finished and you are on time. A short exit ticket: read each question aloud twice, with the four options of question 2, children write on page 5, then check their own ticket on the next slide. Nobody holds the ticket up to the camera. Skip it if you are behind: the quest was the exit for this lesson. |
| 32 | optional | **Check the exit ticket**<br>s-exitans | answers E1, E2, E3 | Read each answer aloud. Children tick their own ticket with a tick. Answers: $3; option 2, $15; $2 and $8. The exit ticket is never part of Mi's Challenge Quest and never shown to the group. |

## 8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

| Chapter | Part | You say (recording voice) | A child watching later |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 0:00 | Opening | *If you are watching the recording: hello! Get a pencil, paper, your workbook, tokens and your Quest Log. Pause or rewind any time.* | Gather your workbook, tokens, pencil, paper and the closed Quest Log. Press play when ready. |
| 2:00 | Remember warm-up | *Recording: join in! Say your answers out loud before I say them. Pause if you need longer.* | Play along. Say each answer out loud before the video does. |
| 8:00 | Sneak-peek puzzles | *Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.* | Pause. Try both puzzles on your own paper. A guess is fine. |
| 12:00 | Explore | *Recording: pause now. Use your tokens to find the cost of one, then of many. Press play when you have tried.* | Pause. Share the tokens to find 1, then count tokens to find many. |
| 22:00 | Mini-lesson | *Recording: keep watching and copy the examples. Rewind if you miss a step.* | Watch and copy the examples. Rewind any step you missed. |
| 31:00 | Brain break | *Recording: stand up and join in! Count in twos and fives with the moves.* | Stand up. Count in twos and fives with the moves, then sit down. |
| 35:00 | Mi's Challenge Quest | *Recording: Mi's Challenge Quest is not recorded. Skip ahead to the workbook. Play the quest as your tutor arranges.* | Skip to the workbook. Play Mi's Challenge Quest live, or at the make-up slot, not from the video. |
| 55:00 | Workbook | *Recording: pause now and do workbook page 1. When you finish a page, press play to check it.* | Pause. Do page 1, then play to check. Then page 2. Then choose a page to try. |
| 80:00 | Talk time | *Recording: pause. Tell a grown-up if Aisha is right, and test it with tokens.* | Pause. Explain your answer to someone at home, and test it with tokens. |
| 84:00 | Look-back | *Recording: pause and fill in your look-back box. You can pass.* | Pause. Fill in the look-back box, or pass. |
| 86:00 | Flex time and goodbye | *Recording: well done for finishing the term! Show a grown-up one thing you found out.* | Show a grown-up your workbook and tell them one thing you found out this term. |

**Do not record these slides.** Pause the recording (or show a blank slide) while the quest is on screen, so the shared recording does not give away the questions or answers: s-quest1 (Mi's Challenge Quest); s-quest2 (The Pearl Cave). Say once to the camera: the recording pauses now for Mi's Challenge Quest, and starts again for the workbook. Pause the recording before you read the first gate.

## 9. Timing check (model, not a stopwatch)

| Part | Allocated | Script | Slow class | Flag |
|---|---|---|---|---|
| opening | 2:00 | 1:33 | 1:45 | ok |
| remember | 6:00 | 4:08 | 4:55 | LOOSE |
| sneak | 4:00 | 2:51 | 3:21 | ok |
| explore | 10:00 | 8:16 | 9:49 | ok |
| mini | 9:00 | 7:29 | 8:50 | ok |
| break | 4:00 | 3:11 | 3:47 | ok |
| exit | 20:00 | 17:09 | 20:10 | OVER (slow class) |
| work | 25:00 | 20:56 | 24:56 | ok |
| talk | 4:00 | 3:20 | 3:57 | ok |
| look | 2:00 | 1:38 | 1:54 | ok |

| | | | | |
|---|---|---|---|---|
| flex | 4:00 | 0:48 | 0:55 | ok |
| **Total** | 90:00 | 71:21 | 84:20 | |

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 6 questions; a middle child needs about 7.5 min for page 1 and 8.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.